Mathematical Biology Summer School

Research Project

Can Stronger Chemotherapy Always Lead to Better Outcomes?

Group members: ______________________________

1. Background

Chemotherapy is one of the most common treatments for cancer. Many chemotherapy drugs are especially effective against rapidly dividing cells. However, a tumor is not made of identical cells. Some tumor cells may divide rapidly and be strongly affected by treatment, while others may divide more slowly and be less affected.

As chemotherapy removes sensitive cells, resistant cells may make up a larger fraction of the remaining tumor. A treatment may therefore shrink the tumor at first while also changing its composition.

In this project, your team will use a mathematical model to investigate how chemotherapy dose and treatment strategy affect two tumor-cell populations.

Main research question

Does increasing the chemotherapy dose always produce a better outcome?

2. The Mathematical Model

Suppose the tumor contains two populations:

- $S(t)$: fast-growing, drug-sensitive tumor cells,
- $R(t)$: slow-growing, drug-resistant tumor cells.

We begin with the model

$$\frac{dS}{dt} = r_S S - c k_S r_S S, \tag{1}$$
$$\frac{dR}{dt} = r_R R - c k_R r_R R. \tag{2}$$

These equations may also be written as

$$\frac{dS}{dt} = r_S(1 - c k_S)S,$$
$$\frac{dR}{dt} = r_R(1 - c k_R)R.$$

The model assumes

$$r_S > r_R \qquad \text{and} \qquad k_S > k_R.$$

Thus, sensitive cells grow faster but are also more strongly affected by chemotherapy.

Meaning of the Parameters

Symbol	Meaning
$S(t)$	Number or volume of sensitive tumor cells at time t
$R(t)$	Number or volume of resistant tumor cells at time t
r_S	Natural growth rate of sensitive cells
r_R	Natural growth rate of resistant cells
k_S	Chemotherapy sensitivity of sensitive cells
k_R	Chemotherapy sensitivity of resistant cells
c	Chemotherapy dose or treatment intensity

Important modeling assumption.
We assume that chemotherapy preferentially kills rapidly dividing cells. In this simplified model, the slower-growing population behaves as the more drug-resistant population. Real drug resistance can arise through many additional biological mechanisms.

3. Quantities to Track

The total tumor size is

$$T(t) = S(t) + R(t).$$

The resistant fraction is

$$f_R(t) = \frac{R(t)}{S(t) + R(t)}.$$

These two quantities answer different questions:

- $T(t)$ tells us whether the overall tumor is growing or shrinking.
- $f_R(t)$ tells us whether resistant cells are becoming more dominant.

A treatment can reduce $T(t)$ while increasing $f_R(t)$.

4. Suggested Research Questions

Your group does not need to answer every question. Choose several connected questions and develop a clear research story.

- Does increasing the chemotherapy dose always reduce the final tumor size?
- What happens in the long-term after treatment?
- At what dose does the sensitive population begin to shrink?
- Can the resistant population continue growing during treatment?
- How does the initial fraction of resistant cells affect treatment success?
- What happens after chemotherapy is stopped?

- Is a short, strong treatment better than a longer, weaker treatment?
- Can two treatments produce the same final tumor size but different resistant fractions?
- Which outcome should be used to judge whether treatment is successful?

You may also develop your own research question.

5. Suggested Investigation Plan

Step 1: Study the Tumor Without Treatment

Set

$$c = 0.$$

Simulate the two cell populations and compare their growth.
Questions to consider:

- Which population grows faster?
- Which population dominates in the absence of treatment?
- How does the total tumor size change?

Step 2: Introduce Chemotherapy

Choose several values of c and simulate the model again.
For each dose, plot

$$S(t), \qquad R(t), \qquad T(t), \qquad f_R(t).$$

Try to identify three possible treatment regimes:

1. both populations grow,
2. sensitive cells shrink but resistant cells grow,
3. both populations shrink.

Extension: Explore Treatment Schedules

Compare different treatment schedules, such as

- no treatment,
- delayed treatment,
- treatment for a fixed time,
- strong treatment for a short time,
- weaker treatment for a longer time.

You may represent a time-dependent treatment by writing $c = c(t)$.
For example,

$$c(t) = \begin{cases} c_0, & 0 \le t \le \tau, \\ 0, & t > \tau. \end{cases}$$

6. Choosing Parameter Values

You may begin with the following example values:

$$r_S = 0.50, \qquad r_R = 0.30, \qquad k_S = 0.8, \qquad k_R = 0.2.$$

Possible initial conditions are

$$S(0) = 0.5, \qquad R(0) = 0.5.$$

These values are dimensionless examples chosen for exploration. They are not intended to represent a particular patient or drug.

7. A Mathematical Hint

The sensitive population shrinks when

$$r_S(1 - ck_S) < 0.$$

Since $r_S > 0$, this occurs when

$$c > \frac{1}{k_S}.$$

Similarly, the resistant population shrinks when

$$c > \frac{1}{k_R}.$$

Because $k_S > k_R$, there may be a range of doses for which sensitive cells decrease while resistant cells continue to grow. Your group should interpret what this means biologically.

8. Final Reminder

There is no single correct answer.
Mathematical modeling is an exploratory process. Different groups may choose different parameter values, investigate different questions, and reach different conclusions. Your goal is to support your conclusions using mathematical reasoning, simulations, and clear biological interpretation.